

AYUSH GOYAL

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Education

University of Southern California

Masters in Computer Science - GPA: 3.9/4.0

June 2024 - Present

Los Angeles, CA, USA

Indian Institute of Technology Delhi

B.Tech. in Mathematics and Computing - CGPA: 9.078/10.0

July 2019 – May 2023

Delhi, India

Teaching Assistant: For an undergraduate course: Machine Intelligence & Learning

Skills

Languages/Tools: Python, C, C++, Java, MATLAB, Verilog, Standard ML, Git, LATEX, Autodesk Inventor

Frameworks/Libraries: PyTorch, TensorFlow, CVXOPT, Keras, Sklearn, Numpy, Pandas, Jupyter

CourseWork: Natural Language Processing, Computer Vision, Machine Learning, Data Mining, Data Structures and Algorithms, Multimedia Systems, Analysis and Design of Algorithms, Quantum Computing and Cryptography, Computer Architecture, Theory of Computation, Operating Systems, Probability and Stochastic Processes, Statistical Methods

Work Experience

Goldman Sachs - Quant Strategist Intern

June 2025 - August 2025

FICC Mortgage Strats & FICC IRP SMM (Summer Analyst)

New York, NY, USA

- Engineered features and trained a Random Forest (scikit-learn) model to predict valuations of buildings in **CMBS**
- Extended the **Spreader** algorithm in **Java** to boost performance under aggressive market conditions
- Investigated spreader slippage vs arrival market price, identified **+\$137K** per \$1mm duration traded opportunity

KAIST - Research Assistant

August 2023 - June 2024

Prof. Jaesik Choi and Research Prof. Nari Kim

Statistical Artificial Intelligence Lab

- Improved XAI performance by **7%** on BERT-base text models using dependency parser-based syntactic units
- Evaluated XAI techniques (**LIME**, **SHAP**, **IG**, **LRP**) on various text models for a Plug and Play XAI project
- Built a pipeline for selecting the optimal explainer for text models, enhancing model trustworthiness

Adobe - Research Intern

June 2022 - August 2022

- Optimized GANs via **Quantization-Aware Training**, improving model size and performance using **PyTorch**
- Analyzed *layer-wise sensitivity* of StyleGAN2 by estimating average Hessian matrix trace via **Hutchinson** algorithm
- Achieved **6.5 times** compression ratio with negligible FID score degradation as compared to the original model

Project Experience

Intent based Counter Speech - Prof. Niladri Chatterjee

- Developed a **multi-modal transformer pipeline** using a **proprietary** dataset for generating counter speech
- Curated a novel dataset by defining parameters for evaluating counter speech derived from the IntentCONAN dataset
- Achieved **88%** top-2 accuracy and **0.59** Pearson correlation for identifying best-intent counter speech

Claim Span Identification (CSI) - Prof. Tanmoy Chakrobarty

- Mitigated the overfitting problem in the SOTA DABERTA model for CSI by focusing on the fully connected layers
- Devised a heuristic post-processing algorithm based on **EDA**-derived claim span properties
- Improved the model's F1 score from 0.834 to 0.842, while also reducing training time by modifying layers

Multi-Threaded Junction Management System - Prof. Ashutosh Rai

- Created C program to optimize memory access & **reduce page faults** by implementing page replacement strategies
- Program handled trace files, monitor memory accesses, and analyze disk operations to enhance memory management

Memory Management and Page Replacement Algorithms - Prof. Ashutosh Rai

- Implemented master thread in C to ensure **liveness** and **prevent starvation** within the concurrent system
- Utilised multithreading, condition variables, and mutex to ensure safety by allowing one train per lane at a time

Leadership & Involvement

Coordinator, Mathematics Society, IIT Delhi: Led team of 30+ members, organized events related to mathematics

Academic Mentor, IIT Delhi: Cleared doubts and assisted in teaching for the Introductory Electrical engineering course